Exact Extension of OEIS A049230 through 27 Steps: Cubic-Orbit Enumeration at 19–22 and Published Partition-Function Data at 23–27

Carlo Corti
Independent researcher
carlo.corti@outlook.com

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Abstract

OEIS A049230 counts rooted n-step self-avoiding walks on the simple cubic lattice having exactly two nearest-neighbour contacts between vertices that are nonconsecutive along the walk. Translations are removed by fixing the initial vertex, whereas rotations, reflections, and reversal remain distinct. At the time of our audit, OEIS A049230 listed a contiguous prefix through $a(18)$. Our literature search identified exact published partition-function coefficients that normalize to $a(23), \ldots, a(27)$. Relative to the OEIS prefix, this left precisely four intervening values undetermined. We compute them as

$$a(19) = 2126459849880, \quad a(20) = 9600694064496,$$
$$a(21) = 43090682243160, \quad a(22) = 192553331372616.$$

The computation enumerates genuinely three-dimensional walks modulo the full 48-element cubic point group and reconstructs the rooted count by the exact identity

$$a(n) = 3\,A033323(n) + 48p^{(3)}_{n,2}.$$

On the computational platform specified below, the frozen campaign used 2411 deterministic recovery units in three segments and records 10232.08 seconds of accumulated per-unit wall-clock time. An independent Java implementation directly reproduced $a(19)$; it differs in software realization and symmetry handling but shares the same exhaustive depth-first, incremental-contact method. The normalized published values at $n = 23, \ldots, 27$ are not outputs of the campaign. The resulting documented range is contiguous for $1 \le n \le 27$; within the evidence assembled in the audit of 26 August 2026, the first unresolved index is $n = 28$.

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1 Introduction and provenance

A self-avoiding walk (SAW) is a nearest-neighbour lattice path that visits no vertex twice. Contact-resolved enumeration is a classical finite-lattice model of polymer–solvent interaction [1]. Nemirovsky, Freed, Ishinabe, and Douglas organized these counts by the number of coordinate dimensions actually spanned and introduced dimension-independent coefficients $p^{(l)}_{n,m}$ [2]. Their framework fixes the symmetry normalization used below.

OEIS A049230 was submitted by N. J. A. Sloane. Petros Hadjicostas clarified its contact interpretation on 3 January 2019, and Sean A. Irvine supplied the displayed terms $a(12), \ldots, a(18)$ on 23 July 2021 [10]. A live inspection on 26 August 2026 still displayed the contiguous range only through $a(18)$. During our literature audit, we identified exact partition-function papers containing later isolated coefficients corresponding to $a(23), \ldots, a(27)$ [5, 6, 7]; the later study of Huang, Hsieh, and Chen independently confirms the exact-enumeration range through 27 steps, while referring to those partition-function sources for fixed-first-step partition polynomials [8]. After exact normalization, this made $n = 19, 20, 21, 22$ the only intervening unresolved indices, rather than every index after 18.

This paper reports an exhaustive computation of precisely those four values. It also normalizes the later published coefficients, thereby documenting a contiguous exact range through 27 without presenting published values as new campaign outputs. The distinction matters both scientifically and for the provenance of the associated b-file.

2 Definition, indexing, and dimensional decomposition

Definition 1. Let $w = (v_0, v_1, \ldots, v_n)$ be an ordered nearest-neighbour walk on $\mathbb{Z}^3$, rooted by $v_0 = 0$. It is self-avoiding if its vertices are distinct. Its number of contacts is

$$m(w) = \#\big\{\{i,j\} : 0 \le i < j \le n,\ j-i > 1,\ \|v_i - v_j\|_1 = 1\big\}.$$

The value $a(n) = A049230(n)$ is the number of such walks with $m(w) = 2$. The vertex order is retained, so reversal is distinct; no rotation or reflection quotient is taken in $a(n)$.

The walk has n steps and $N = n+1$ monomers. This conversion is essential when partition polynomials are indexed by monomer count. In this paper,

$$Z_N(x) = \sum_{w \in \mathcal{W}^{(3)}_{N-1}} x^{m(w)},$$

where $\mathcal{W}^{(3)}_{N-1}$ is the set of rooted ordered $(N-1)$-step simple-cubic SAWs with all six possible first-step directions. Let $\widehat{\mathcal{W}}^{(3)}_{N-1}$ be the subset whose first step is $+x$, and write

$$\widehat{Z}_N(x) = \sum_{w \in \widehat{\mathcal{W}}^{(3)}_{N-1}} x^{m(w)}.$$

The cubic point group acts transitively on the six first-step directions without changing contacts, so

$$Z_N(x) = 6\widehat{Z}_N(x), \qquad a(n) = [x^2]Z_{n+1}(x) = 6[x^2]\widehat{Z}_{n+1}(x).$$

Definition 2. For $d \ge 1$, let $C_{n,m}(d)$ be the number of rooted ordered n-step SAWs on $\mathbb{Z}^d$ with exactly m contacts, with all $2d$ first-step directions retained and with reversal distinct. For $1 \le l \le n$, let $p^{(l)}_{n,m}$ be the number of orbits of rooted ordered n-step SAWs on $\mathbb{Z}^l$ with exactly m contacts whose step vectors use all l coordinate axes, under the order-$2^l l!$ hyperoctahedral group of signed coordinate permutations of $\mathbb{Z}^l$. These walks are called genuinely l-dimensional; lower-dimensional walks are excluded from $p^{(l)}_{n,m}$. Embedding $\mathbb{Z}^l$ in any fixed coordinate l-subspace of $\mathbb{Z}^d$ gives a bijection with the genuinely l-dimensional walks in that subspace. Thus the orbit count is independent of the ambient dimension, while the choice of subspace contributes the factor $\binom{d}{l}$ below.

Nemirovsky et al. give the dimensional decomposition

$$C_{n,m}(d) = \sum_{l=1}^{\min(n,d)} 2^l l! \binom{d}{l} p^{(l)}_{n,m}. \tag{1}$$

This is Eq. (5) on p. 1090 of their paper [2].

Proposition 3. *For every $n \geq 1$,*

$$a(n) = 3\,A033323(n) + 48p^{(3)}_{n,2}. \tag{2}$$

Proof. A one-dimensional SAW is a monotone straight walk, so it has no contact between nonconsecutive vertices and $p^{(1)}_{n,2} = 0$. OEIS A033323 uses the same step index, rooting, vertex order, and contact convention on the square lattice; hence

$$A033323(n) = C_{n,2}(2) = 2^2 2!\, p^{(2)}_{n,2} = 8p^{(2)}_{n,2}.$$

Specializing Eq. (1) to $d = 3$ gives

$$C_{n,2}(3) = 24p^{(2)}_{n,2} + 48p^{(3)}_{n,2} = 3A033323(n) + 48p^{(3)}_{n,2}.$$

Since $C_{n,2}(3) = a(n)$, the result follows. □

The derivation is structural, not a numerical fit, and the two input tables were joined on the mathematical index n. In particular, the planar inputs for $n = 19, \ldots, 22$ originate in Table B1, pp. 4738–4739, of Bennett-Wood et al. (Table 4, pp. 26–28, in the corresponding preprint). Its column $c_n(2)/4$ is $A033323(n)/4$ in the present notation, so multiplication by four recovers $A033323(n)$ [4, 11]. For $n = 19, \ldots, 22$, the four recovered values coincide with the corresponding OEIS A033323 terms. The OEIS record credits Hadjicostas with the source interpretation, Irvine with terms $a(20), \ldots, a(26)$, and Corti with $a(27), \ldots, a(29)$; the primary numerical provenance of all planar values used here remains the Bennett-Wood et al. table.

3 Symmetry and stabilizers

The full octahedral group O_h, equivalently the three-dimensional hyperoctahedral group, consists of the $2^3 3! = 48$ signed coordinate permutations. It acts on a rooted walk coordinatewise. Its action is free on a genuinely three-dimensional ordered walk: an element fixing the ordered walk fixes every vertex, hence every step vector; because the walk contains three linearly independent step vectors, the element is the identity. Consequently every genuinely three-dimensional orbit has size 48.

Planar walks behave differently. Reflection through their supporting coordinate plane fixes every vertex, so they have a nontrivial stabilizer in the three-dimensional group. Their contribution is therefore not obtained by blindly dividing the full count by 48. Instead A033323 supplies the oriented count in one abstract square lattice and the factor 3 in Eq. (2) chooses one of the three coordinate planes. This separation is why the reconstruction treats planar and genuinely three-dimensional strata independently.

Proposition 4. *Every genuinely three-dimensional orbit has exactly one representative that satisfies all of the following conditions:*

1. *the first step is $+x$;*

2. the first step along the y-axis is $+y$, and no step along the z-axis occurs before it;

3. the first step along the z-axis is $+z$.

Proof. The first condition chooses one of the six signed directions of the first step. After it is fixed, the residual group has order eight. Let t_y and t_z be the first times at which the image uses the y- and z-axes; genuine three-dimensionality makes both finite and distinct. Of the identity and the transverse-axis transposition, exactly one gives $t_y < t_z$. The two remaining sign freedoms then independently make the first y step $+y$ and the first z step $+z$. Hence the transverse-axis ordering removes the residual swap, the two sign conditions remove the residual reflections, and all eight elements of the stabilizer of $+x$ are resolved uniquely. □

The frozen production search implements precisely these conditions: it fixes the first step to $+x$, rejects a z step until a positive first y step has occurred, rejects a negative first step along either transverse axis, and counts a state only after z has been used. Proposition 4 therefore implements the factor 48 in Eq. (2); reversal is never identified.

The corresponding partial-walk admissibility predicate is closed under taking prefixes: a prefix of a canonical walk cannot have the wrong first step, use z before y, or have a negative first use of either transverse axis; an axis not yet used creates no violation. Since the contact count is also nondecreasing, every canonical target walk of length at least 10 has an admissible length-10 prefix with at most two contacts. The predicate may therefore be enforced incrementally, and the depth-10 frontier is exhaustive.

4 Exhaustive enumeration

The enumerator is a single C17/OpenMP program. For a maximum length n, each worker maintains a private byte-valued occupancy grid of side $2n + 3$, padded by one lattice layer beyond every reachable coordinate. It fixes the first step to $+x$ as in Proposition 4, then extends the walk in the deterministic direction order $+x, -x, +y, -y, +z, -z$. A candidate vertex is rejected if it is already occupied. Otherwise the number of newly created nonbonded contacts is

$$\#\{\text{occupied nearest neighbours of the candidate}\} - 1,$$

because the predecessor is the unique occupied bonded neighbour. Contacts cannot disappear as a branch grows, so any branch exceeding two contacts is pruned. At every depth n, a genuinely three-dimensional state with exactly two contacts contributes one to $p^{(3)}_{n,2}$.

The implementation, build recipe, campaign outputs, and validation materials described in this paper form the accompanying project package [13].

The search tree was split at depth 10 into 154271 deterministic prefix tasks, where a task is one admissible canonical partial walk together with its contact and transverse-axis state. The tasks were grouped in generation order into 2411 recovery units, each a consecutive range of at most 64 tasks. A segment is one operator invocation that completes whole recovery units until the configured time target or the final unit is reached. The manifest is the durable table containing the frozen configuration, segment state, and one ordered record per completed unit, including its task interval, elapsed time, and per-depth counts.

Each OpenMP worker used a private occupancy grid and private `uint64_t` counters. A critical merge performed checked additions and failed on overflow; the program's structural cap at $n = 29$ is justified by the nonbacktracking orbit bound $p^{(3)}_{n,2} \le 5^{n-1}/8$, which is below the 64-bit limit through that cap. The manifest bound the frozen source and build identity to the configuration, recorded every completed unit exactly once, and published state by no-clobber atomic replacement. The verifier regenerated the depth-10 frontier and required its task count to equal the manifest's 154271 before aggregating all unit records. Interrupted execution resumes only at a recovery-unit boundary, so segmentation changes neither the search space nor the arithmetic result.

The official configuration used 8 OpenMP threads, split depth 10, unit size 64, and a 4500-second soft segment target. The target computer had an AMD Ryzen 9 7940HS processor (8 cores, 16 logical processors) and 64 GB DDR5 installed RAM; the operating system reported 61 GiB total memory. It ran Linux Mint 22.3 with 64-bit Linux kernel 6.17.0-42-generic. The executable was built with GCC 13.3.0 and GNU libgomp. The relevant Makefile flags were

```
-O3 -flto -funroll-loops -fomit-frame-pointer -march=native
-mtune=native -std=c17 -Wall -Wextra -Wpedantic -Wshadow
-Wconversion -Wdouble-promotion -Wformat=2 -fopenmp -lm
```

The campaign was run against that frozen software and environment. It completed all 2411 units in three normally terminating segments. Their individually rounded compute times were 4500, 4501, and 1231 seconds. These displayed values sum to 10232 seconds. The target is checked only before starting the next complete recovery unit, which explains the small overshoot of the first two segments. These quantities are wall-clock compute time, not CPU time or campaign calendar time; manual pauses between segments are excluded. They are specific to the stated hardware, software, compiler flags, and eight-thread configuration.

5 Validation and independence boundary

Validation addressed the mathematical predicate, symmetry normalization, indexing, persistence, and implementation separately.

1. The frozen executable reproduced $p^{(3)}_{n,2}$ for $1 \leq n \leq 18$. Reconstruction by Eq. (2) reproduced the previous A049230 prefix byte for byte.

2. A direct oriented oracle exhaustively enumerated the full rooted target set through $n = 8$ without using the production canonical-orientation predicate. A separate signed-permutation oracle partitioned the same full target sets into orbits, explicitly generated all 48 images of every orbit, and verified both orbit size 48 and exactly one representative satisfying Proposition 4.

3. Crippen's independent Table I gives the number $N_2(n+1)$ of cubic symmetry classes with two contacts for $5 \leq n \leq 11$. In the present notation it checks

$$N_2(n+1) = p^{(2)}_{n,2} + p^{(3)}_{n,2} = \frac{a(n) + 3A033323(n)}{48},$$

   after reconciling rooting, reversal, and symmetry conventions [3]. For $n = 5, \ldots, 11$, the common values are $8, 52, 270, 1446, 7578, 38473, 191154$.

4. The primary Shirai–Sakumichi attachment tabulates endpoint-constrained counts $W_{n,m}(r)$ for endpoints $(r, 0, 0)$. Strict aggregation at $m = 2$ gave $\sum_r W_{19,2}(r) = 6927757800$ and $\sum_r W_{20,2}(r) = 28720091852$; multiplying by the six axial directions gives exact subsets 41566546800 and 172320551112 [9]. These marginals test depth-19 and depth-20 coverage but are not the unrestricted totals.

5. The completed manifest passed strict continuity, integrity, source identity, task-coverage, and final-state checks. Thread-count and split-mode equivalence tests, compiler warnings, two compiler families, static analyzers, address/undefined-behaviour sanitizers, and Valgrind checks were recorded before the campaign. Unavailable ThreadSanitizer and system-call tracing environments were recorded as unavailable, not as passes.

After the result was frozen, a separate Java verifier based on Sean A. Irvine's jOEIS class `ExactContactsWalker` fixed the first step to $+x$, directly enumerated that complete rooted sector, and restored the six first-step directions by exact weighting. It returned 2126459849880 at $n = 19$. It does not use the production cubic-orbit selector, so this run also checks the factor-48 normalization at that index. The verifier was bound to a fixed jOEIS source revision recorded in the validation package. The upstream sources and local harness were verified, compiled in a fresh temporary directory, and did not read the C campaign, A033323, the Python derivator, or the b-file [12].

This is an *independent software reproduction*, not an independent mathematical method. The two implementations differ in author, language, coordinate and occupancy representation, symmetry accounting, parallel decomposition, compiler chain, and result path. Both nevertheless perform exhaustive depth-first SAW enumeration with incremental contact accumulation. The validation claim is deliberately limited to $a(19)$.

6 Results and provenance boundary

The computed orbit counts and exact reconstructions are shown in Table 1. The planar term is small but was retained exactly; no rounded or row-position arithmetic entered the derivation.

Table 1: New campaign results. The final column is evaluated exactly by Eq. (2).

n	$p^{(3)}_{n,2}$	$3A033323(n)$	$a(n)$
19	44296694918	218493816	2126459849880
20	200002700951	564418848	9600694064496
21	897692330931	1450358472	43090682243160
22	4011450432806	3710597928	192553331372616

Computational Result 5. *Subject to correct execution of the frozen C17/OpenMP enumerator and the integrity checks on its completed manifest, the exhaustive campaign determines the four A049230 values in Table 1. The independently implemented jOEIS run reproduces the first of these values.*

The later rows have a different provenance. Lee, Kim, and Lee print the full $N = 24$ contact inventory with all first-step directions in Table I on p. 2; its contact-number row $K = 2$ is 856234452257592, hence $a(23)$ directly [5]. Hsieh, Chen, and Hu print fixed-first-step polynomials, denoted here by $\widehat{Z}_N$: $\widehat{Z}_{27}$ and $\widehat{Z}_{26}$ on p. 4 and $\widehat{Z}_{25}$ on p. 5 of their EPJ paper, and $\widehat{Z}_{28}$ in Sec. 3.1 on p. 6 of their CPC paper. Their x^2 coefficients for $\widehat{Z}_{25}$, $\widehat{Z}_{26}$, $\widehat{Z}_{27}$, and $\widehat{Z}_{28}$ are, respectively,

$$632297080298076, 2790018300205392, 12274198909485704, 53807607703672080.$$

Multiplication by six and conversion $n = N - 1$ give $a(24), \ldots, a(27)$ [6, 7]. Huang, Hsieh, and Chen later use exact end-to-end polynomials through 27 steps and explicitly direct readers to these

Table 2: The completed exact chain. "Campaign" means exhaustive output of the computation reported here; "published" means normalized primary-source data, not a campaign output.

n	$a(n)$	Evidence class
19	2126459849880	campaign; Java reproduction
20	9600694064496	campaign
21	43090682243160	campaign
22	192553331372616	campaign
23	856234452257592	published, Lee–Kim–Lee
24	3793782481788456	published, $6[x^2]\widehat{Z}_{25}$
25	16740109801232352	published, $6[x^2]\widehat{Z}_{26}$
26	73645193456914224	published, $6[x^2]\widehat{Z}_{27}$
27	322845646222032480	published, $6[x^2]\widehat{Z}_{28}$

earlier sources for the corresponding fixed-first-step polynomials [8, Appendix A]. Table 2 keeps the published data visibly separate from the campaign.

The canonical b-file contains exactly the continuous index range 1 through 27. Rows 1–18 are the unchanged prior prefix, rows 19–22 are the campaign reconstructions, and rows 23–27 are the normalized published coefficients. A dated literature and exact-data audit completed on 26 August 2026 searched the A-number, the notations $C_{n,2}(3)$ and $p^{(3)}_{n,2}$, cubic contact classes, exact $Z_N(x)$ coefficients, related OEIS records, forward and backward citations of the primary papers, supplementary data, and exact integer fingerprints. An admissible datum was required to be a full unrestricted rooted two-contact count with enough convention information for exact normalization. No such coefficient was found for $n = 28$ or above. Accordingly, within the evidence assembled in this audit, 27 is the documented endpoint and 28 the first unresolved index. This bounded statement neither claims universal nonexistence nor authorizes a recomputation of a later row.

7 Finite numerical observations

Two exact finite patterns provide useful checks but are not asymptotic claims. First, Eq. (2) implies divisibility by 24: both $3A033323(n) = 24p^{(2)}_{n,2}$ and $48p^{(3)}_{n,2}$ are multiples of 24. Every b-file row through 27 satisfies this condition. The stronger normalization check

$$\frac{a(n) - 3A033323(n)}{48} \in \mathbb{Z}_{\geq 0}$$

also passes for each published row $n = 23, \ldots, 27$, giving respectively

$$17838020877573, \quad 79036635370667, \quad 348751024443185,$$
$$1534271679749333, \quad 6725942965307646.$$

These are derived orbit counts used as transcription checks, not additional campaign outputs. The planar inputs in these five checks are $A033323(23), \ldots, A033323(27) =$

$$3150044696, \quad 7994665480, \quad 20209319824, \quad 50942982080, \quad 127962421824,$$

respectively; they are the Bennett-Wood table values after the conversion specified above and coincide with the corresponding OEIS A033323 terms [4, 11].

Second, over the newly computed block the planar share $3A033323(n)/a(n)$ decreases from 0.010275% at $n = 19$ to 0.001927% at $n = 22$ (the intermediate values are 0.005879% and 0.003366%). The successive ratios $a(n)/a(n-1)$ also decrease throughout the currently completed block, from 4.539875 at $n = 19$ to 4.383798 at $n = 27$. These are descriptive observations on a short exact range. They neither establish monotonicity beyond 27 nor estimate a critical exponent or growth constant. No new conjecture is proposed.

8 Conclusion

The literature audit first identified $n = 19, 20, 21, 22$ as the only unresolved indices intervening between the OEIS prefix through $a(18)$ and the normalized published values at $n = 23, \ldots, 27$. The exhaustive orbit-reduced computation reported here determines those four values. Exact dimensional reconstruction, old-prefix replay, small direct and explicit-orbit oracles, contact-class and axial checks, manifest validation, and a separate direct Java reproduction at $n = 19$ give complementary evidence. The boundary is therefore precise: the new campaign ends at 22, the documented sequence ends at 27, and, within the evidence assembled in the audit of 26 August 2026, 28 remains unresolved.

Data availability

The source, Makefile, frozen campaign manifest and result table, derived sequence table, canonical b-file, validators, independent jOEIS harness, completion report, manuscript source, and rendered paper are publicly available in the synchronized GitHub repository and Zenodo archive listed in the References.

AI use disclosure

Large language models were used as research, code-review, validation-planning, artifact-auditing, and drafting assistants during the A049230 project. The preserved substantive records identify OpenAI Codex and review systems from the ChatGPT, Claude, DeepSeek, Gemini, Grok, Kimi, Lumo, Meta, MiniMax, Qwen, and Z families. They are named at family level where exact model-version metadata were not consistently available; no missing version identifier has been reconstructed from inference.

Mathematical analysis. LLMs assisted in restating the dimensional decomposition, challenging the cubic-orbit and planar-stabilizer arguments, locating step/monomer and first-step normalization risks, and identifying exact checks from contact polynomials and related sequences. Accepted statements were checked against the OEIS snapshots, primary papers, exact arithmetic, the direct rooted oracle, and the explicit 48-image orbit oracle. Conversational assertions were not treated as mathematical evidence. During external manuscript review, several independent reviews identified the omitted transverse-axis ordering condition in the written canonical rule. The correction in Proposition 4 was accepted only after direct reconciliation with the frozen C source and the exhaustive small-n orbit oracle.

Algorithm and code. OpenAI ChatGPT/Codex and Claude-family systems supplied the final corrective code reviews; systems from the other named families contributed substantive review artifacts at earlier stages. Their work included static or dynamic code review, adversarial test design, persistence analysis, parser and output-boundary review, and artifact inspection. The final numerical claims rest on the frozen C17 program and build recipe, the completed manifest and result table, known-term replay, cross-thread and split-depth tests, bounded direct and orbit oracles, axial-data checks, the arithmetic derivator, and the separate jOEIS reproduction at $n = 19$. Shared-backend checks and unavailable tool evidence were not promoted to independent proofs.

Project workflow. AI systems assisted in organizing bounded review rounds, comparing findings, maintaining evidence boundaries, and preparing operator-facing runbooks. They did not launch or operate the official computation. Carlo Corti authorized, started, monitored, and completed the campaign and made the scientific and editorial decisions.

Manuscript. Codex assisted with literature research, drafting, claim-to-source auditing, arithmetic consistency checks, and LaTeX diagnostics. Every numerical result retained in the final manuscript was checked against a result table, the canonical b-file, the completed manifest, validator output, exact arithmetic, or a cited primary source. The final claims depend on those artifacts and sources, not on LLM conversation.

Carlo Corti takes full scientific responsibility for the mathematical statements, algorithm, implementation, numerical results, validation boundaries, and conclusions of this paper.

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